Milestone D: Membrane Tool Redesign - Delivery Report

Date: 28 January 2026

Milestone: Milestone D - Membrane Tool Redesign

Status: ✓ Completed

Preview URL: science-canvas-creator.vercel.app

Executive Summary

Milestone D has been successfully completed. The membrane tool has been redesigned from a dialog-based workflow to an integrated path-based tool. Users now draw membrane paths using the familiar pen tool interface, then configure and edit membranes through the Properties panel. All milestone deliverables have been implemented and all manual tests are passing.

Deliverables Completed

Path-Based Membrane Creation

Drawing Workflow:

1. Select the membrane tool from the toolbar
2. Draw a custom path using pen tool interaction (click to add points, drag to create curves)
3. Complete the path by double-clicking or connecting to the first point to close the loop
4. Default phospholipid icons automatically populate along the path with proper alignment and spacing

Path Types Supported:

- Open paths with distinct start and end points
- Closed paths that form continuous loops for cell membranes

Properties Panel Integration

Membrane Configuration:

When a membrane object is selected, the Properties panel displays a "Membrane" section with controls for:

- **Icon Selection:** Choose from membrane-specific organelles (channels, receptors, pumps, carriers)
- **Icon Size:** Adjust the size of all icons along the path
- **Spacing:** Control the distance between icons
- **Orientation Offset:** Rotate icons by a specific angle
- **Rotate to Follow Path:** Toggle automatic alignment with path direction
- **Double-Sided (Bilayer):** Display icons on both sides of the path for realistic membrane visualization

All changes apply immediately to the membrane object.

Two-Tab Edit Mode

Entering Edit Mode:

Double-click any membrane object to enter edit mode, which provides two tabs:

Line Tab (Path Editing):

- Edit the guiding bezier path using the same controls as the pen tool
- Move anchor points by dragging them
- Adjust curve handles to reshape the path
- Icons automatically reposition and reorient as the path changes
- All changes update in real-time

Icon Tab (Icon Editing):

- Select individual icons within the membrane
- Move, rotate, or delete selected icons
- Add new icons from the icon library that automatically integrate into the membrane structure
- Search and filter membrane-specific icons
- Use the "membrane only" checkbox to show only appropriate organelles

Membrane Icon Library

Curated Selection:

- Icon library filtered to show only membrane-appropriate organelles
- Search by name or browse by category
- Admin-controlled designation ensures only suitable icons appear
- Preview icons before adding to the membrane

Admin Panel Configuration

Global Default Settings:

New "Membrane Tool Defaults" section in the Admin panel allows administrators to configure:

- Default membrane icon (e.g., phospholipid)
- Default icon size, spacing, and orientation
- Default rotation behavior (follow path or fixed angle)
- Default bilayer rendering (single or double-sided)

These settings apply to all new membranes created by any user, ensuring consistency across the platform.

Testing Results

All milestone manual tests have been executed and are passing:

- ✓ **Icons align correctly on curved paths** - Icons maintain proper orientation on complex curves
- ✓ **Double-click edit mode matches pen tool behavior** - Consistent editing experience
- ✓ **Icon resizing feels smooth and proportional** - Real-time updates with proper scaling
- ✓ **Properties panel is more discoverable** - Integrated controls more intuitive than previous dialog

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Additional Improvements

Beyond the scope of Milestone D, the following enhancements were delivered:

Undo/Redo System Improvements

- Resolved edge cases where multiple undo actions were required for a single change
- Related changes now batch into single undo steps
- Improved history stack reliability

Pen Tool Enhancements

- Added ability to close paths by connecting the last point to the first point during drawing
- New "Close Path" button in edit mode converts open paths to closed paths
- Improves workflow for creating membrane loops

Conclusion

Milestone D has been delivered in full. The redesigned membrane tool successfully addresses all limitations from the original implementation:

- ✓ Icon library filtered to membrane-specific organelles only
- ✓ Icons properly aligned with correct directional orientation
- ✓ Paths can be modified after creation through double-click edit mode
- ✓ Individual icons can be swapped and adjusted dynamically
- ✓ New icons integrate seamlessly into existing membranes
